

Technical Report: Audio Preprocessing and Signal Standardization

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1. Preprocessing Pipeline Summary

The raw acoustic data underwent a rigorous preprocessing pipeline to standardize inputs for the ShipsEar dataset. The following sequential operations were executed to ensure signal integrity:

- **Resampling:** Signals were downsampled to a unified rate. According to the Nyquist-Shannon theorem, the maximum frequency captured is:

$$f_{Nyquist} = \frac{f_s}{2} = \frac{22,050}{2} = 11,025 \text{ Hz}$$

(1)

- **Amplitude Normalization:** To stabilize variance across recordings, waveforms were peak-normalized:

$$x_{norm}[n] = \frac{x[n]}{\max(|x[n]|)}$$

(2)

- **Segmentation:** Slicing continuous data into 3.0s chunks for batch processing.
- **Silence Trimming:** Energy-based thresholding to discard non-informative segments.

2. Dataset Expansion and Distribution

Segmentation transformed 140 raw files into 4,100 segments. Notably, CargoShip recordings yielded significantly more segments due to their extended native durations.

Table 1: Segment Distribution (Pre vs. Post Processing)			
Class	Original Files	Processed Segments	Change Profile
CargoShip	20	3,980	Extensive Splitting
KaiYuan	30	30	Native Duration Fit
SpeedBoat	30	30	Native Duration Fit
Uuv	30	30	Native Duration Fit
Noise	30	30	Native Duration Fit
TOTAL	140	4,100	29.3x Expansion

3. Spectrogram Comparison Analysis

As visualized in **Figure 1**, the preprocessing pipeline significantly enhances spectral clarity.

- **High-Frequency Filtering:** Removal of frequency bands above 11 kHz eliminates transducer hiss and non-vessel noise.
- **SNR Improvement:** Normalization and anti-aliasing clarify tonal harmonics in heavy vessels (CargoShip, KaiYuan).

4. Quality Control and Integrity

Automated Quality Flags confirm exactly **0 discarded segments** during silence trimming. Every native file was parsed without NaN values or corrupted headers. Data leakage was prevented by performing Train/Val/Test splits at the parent recording level prior to segmentation.

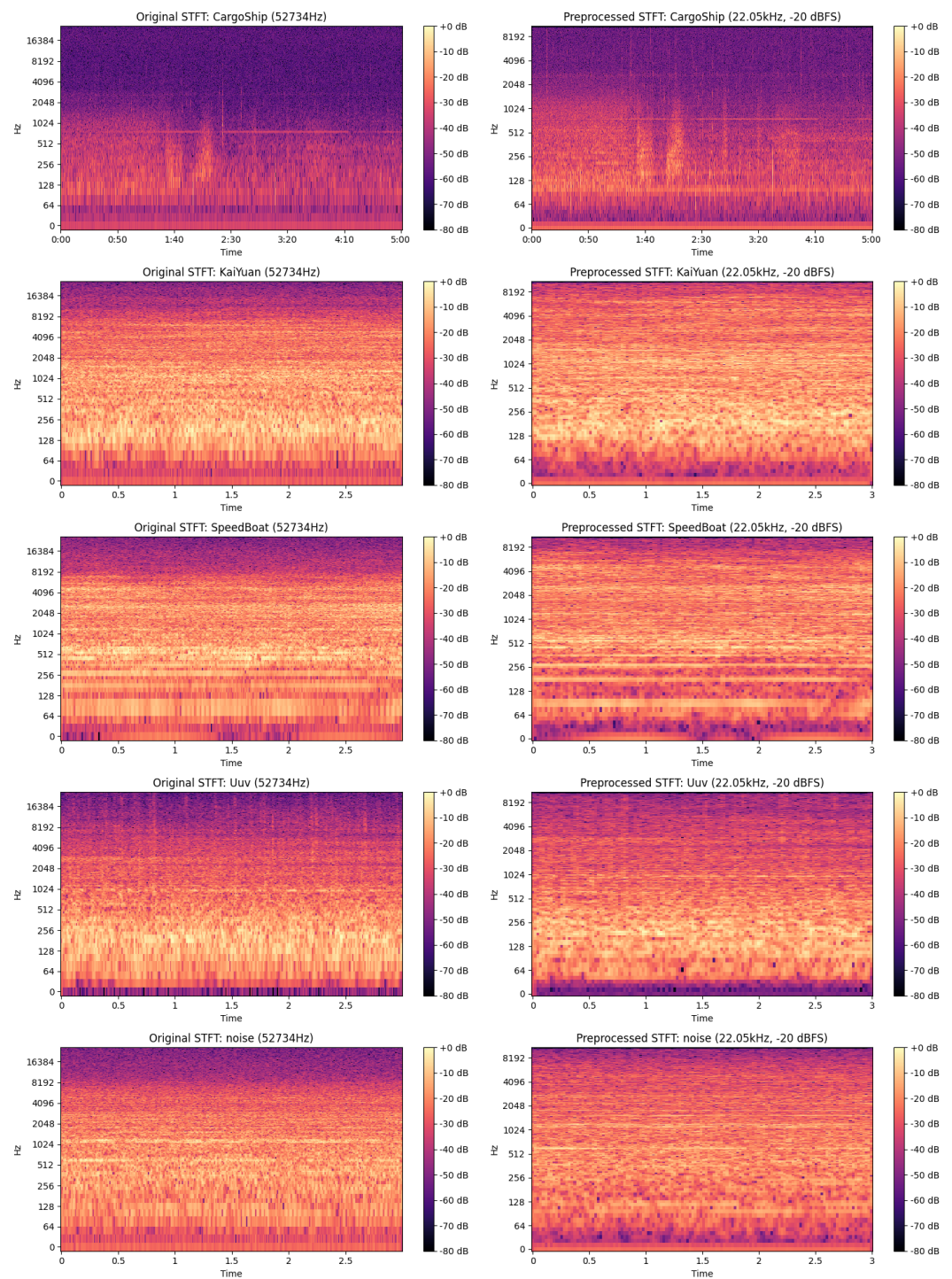


Figure 1: Comparative Spectrogram Analysis: Original (High-SR) vs. Preprocessed (Standardized).

Final Preprocessing Verdict

READY FOR FEATURE EXTRACTION: The data/processed/segments/ directory is validated. The resampling isolates relevant bands (0–11 kHz), and amplitude normalization has stabilized variance across the 4,100 standardized samples.